Roll No. ....

TPE–501

B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

PETROLEUM GEOPHYSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the major role of Geophysics in Petroleum Industry ? What are the various earth properties are measured in Geophysics ? (CO1)

(b) How is Geophysics used for imaging the subsurface structure ? Explain contour map. What are the different recorded parameters used in this process ? (CO1)

(c) Write the importance of passive and active methods of Geophysics. What is the objective of geophysical investigations ? (CO1)

P. T. O.

2. (a) How many geophysical methods are available for oil exploration ? Explain the Gravity and magnetic survey method in detail. (CO2)

   (b) Explain with neat diagram Electromagnetic methods of geophysical survey. Where this survey is useful ? (CO2)

   (c) Explain electrical and well logging methods for oil exploration. What is the data format structure of Seismic data ? (CO3)

3. (a) What is seismic method for exploration ? What are the parameters measured in this method ? Why Seismic is widely accepted in Exploration ? (CO3)

   (b) What is the principle of Geophones ? How to avoid phenomenon of resonance while recording the data ? (CO3)

   (c) Explain Common Depth Point (CDP) Reflection Technique by drawing the figure. Write formula for foldage. (CO3)

4. (a) Explain seismic reflection prospecting methodology by drawing a figure. Write the formula for reflection coefficient. Why is it required ? (CO4)

   (b) What are the various corrections applied in reflection and refraction during the seismic data acquisition ? (CO4)

   (c) Explain NMO correction. Why is it required ? What is diffraction ? Explain with figure. (CO4)

5. (a) Explain Convolution model of seismic trace. How is the seismic trace generated by this method ? (CO5)

   (b) How VSP data is recorded ? Why is it required ? What is the application of VSP data ? (CO5)

   (c) How T-D relationship curve is generated ? Why is it important ? How well to seismic tie is performed ? (CO5)

Roll No. ....

TPE–502

B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

WELL LOGGING AND FORMATION EVALUATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

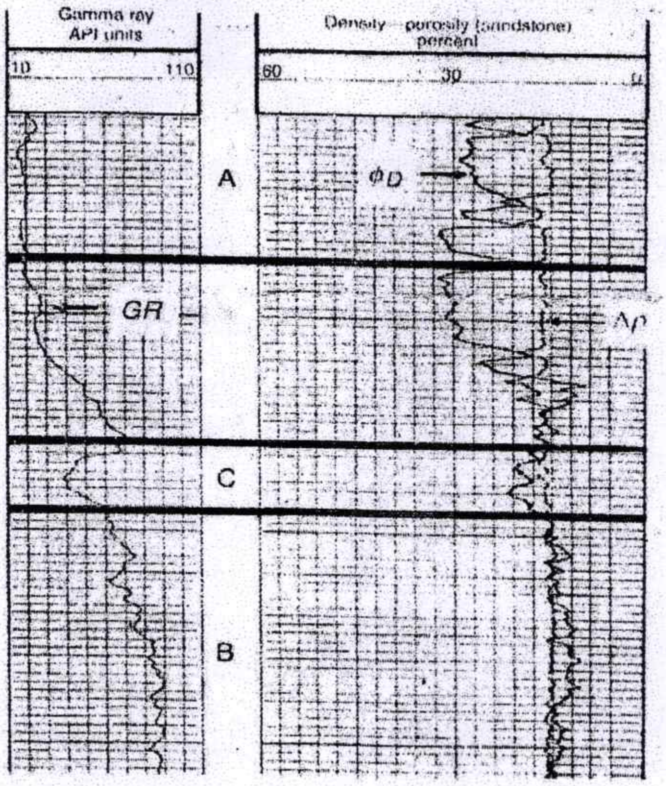
1. (a) Explain various type of well logging available in Oil and Gas field operations. Write the name of 4th generation logging tools in the industry and its applications. (CO1)

   (b) What are the various tools for measurement of lithology, porosity, permeability, hydrocarbon saturation and hydrocarbon type ? Explain in detail. (CO2)

   (c) Explain vertical resolution and depth of investigation of various logging tools. Explain invasion process and Its effects in detail during drilling of well with diagram. (CO1, CO2)

2. (a) (i) Explain the principle of GR tool in details.

(ii) From the given figure explain all three zones A, B and C based on GR and density porosity measurements. (CO2, CO3)



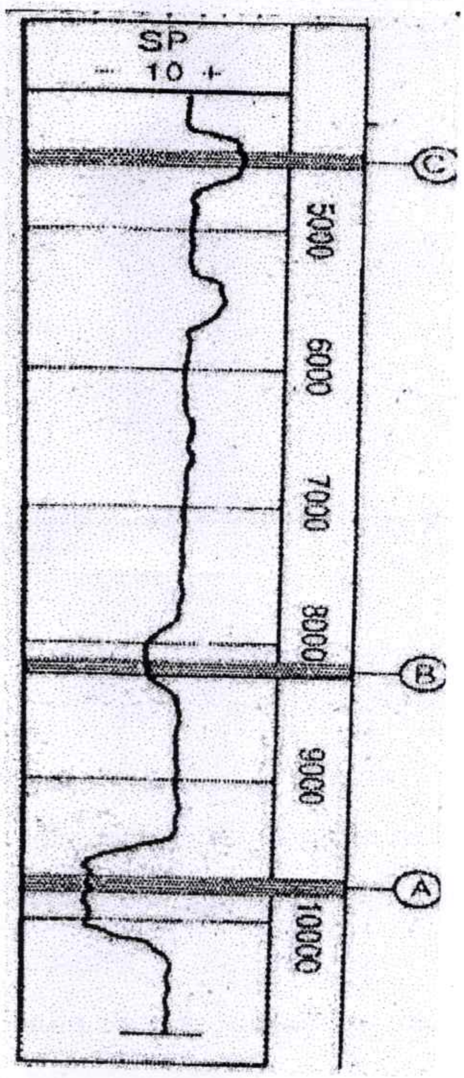
(b) Write the principle of Caliper tool for measurement of hole diameter with diagram. How porous and permeable zone can be identified with caliper log ? Explain with figure. (CO2)

(c) Write formula for volume of shale estimation using GR Log. Solve for V_{shale} if $GR_{Log} = 75$, $GR_{shale} = 120$ and $GR_{Clean} = 30$. (CO3)

3. (a) (i) What is SP log ? (CO2, CO3)

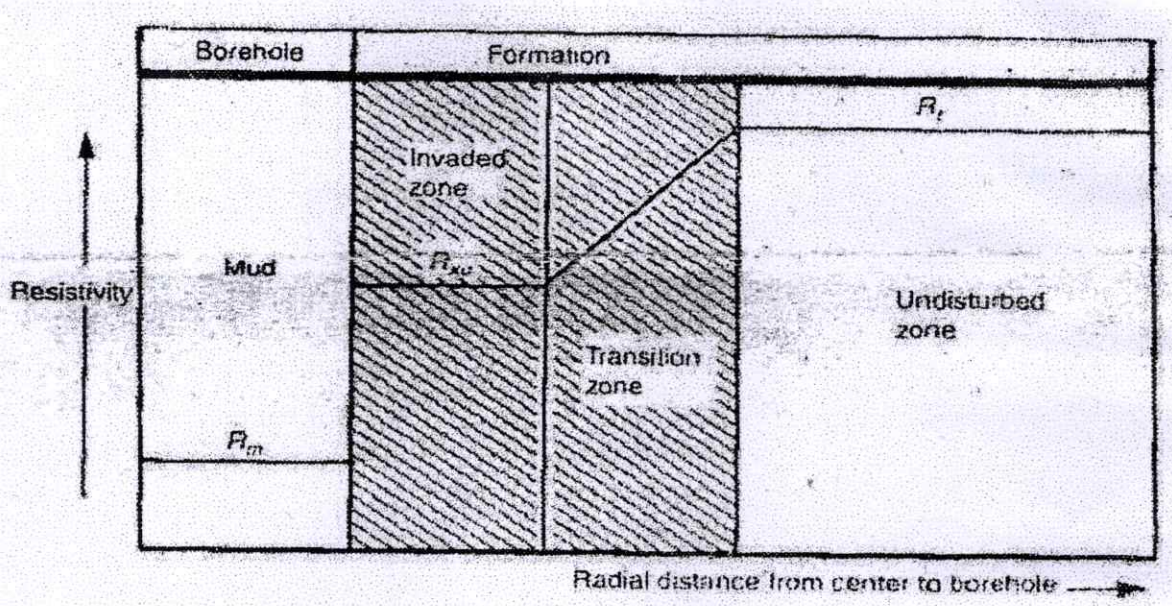
(ii) How Michael Faraday demonstrated SP Log generation in his electrolysis experiments ?

(iii) From the figure explain normal and reverse SP and the reason of it.



(b) (i) Explain Ohm's Law. Write the equation for equivalent resistance for resistors in parallel and in series combination. Write formula for Resistivity. (CO3, CO4)

(ii) Explain from the figure the resistivity behavior in different zones. (CO3, CO4)



(c) Classify the different tools for measurement of resistivity in oil base and water base mud system. Explain their operating principle with figures. (CO4)

4. (a) Explain the principle of measurement of formation density from Density tool. What are the uses of density log ? (CO3)

(b) (i) Draw a neat diagram of neutron tool in the borehole for recording Neutron Log and explain its principle for measurement. (CO3)

(ii) What are the uses of Neutron log ?

(iii) From the figure write the value of GR, RHOB, NPHI at the following depths :

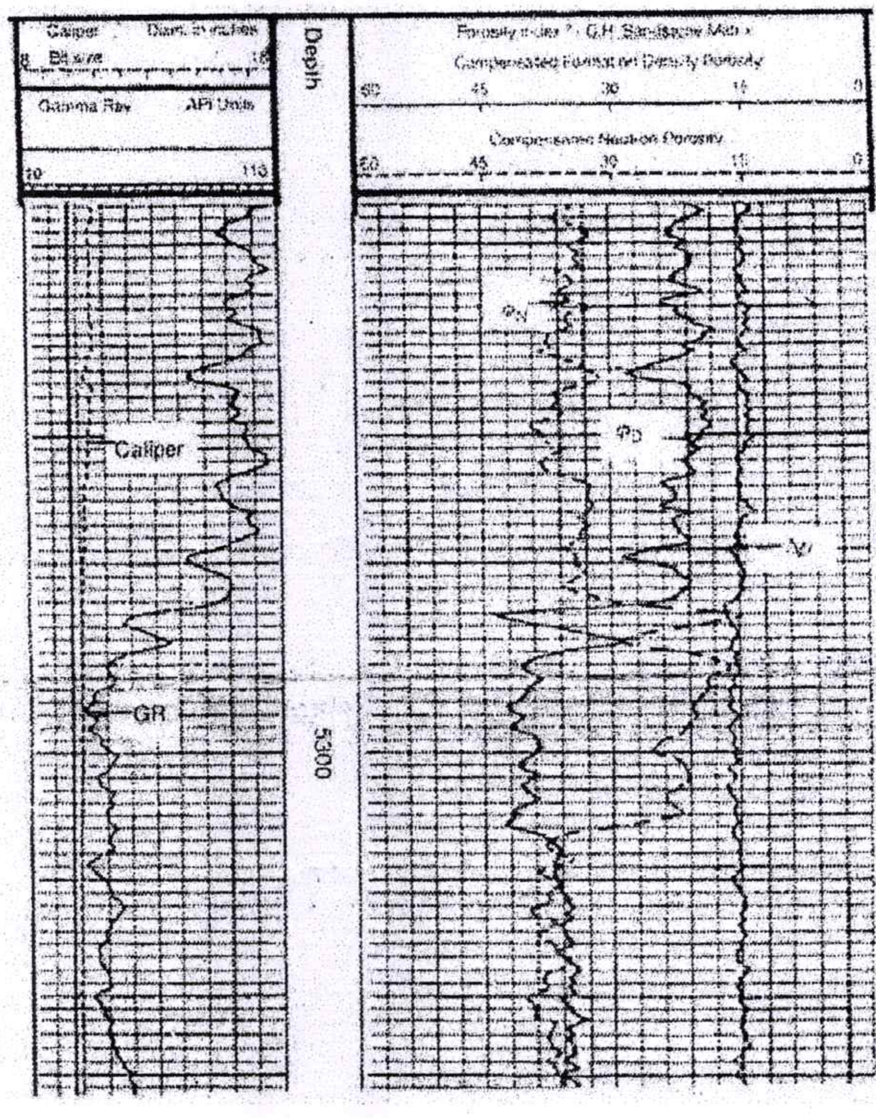
5100 m

5200 m

5300 m

5400 m

Each division in vertical scale is of 10 mt.



(c) Write the formula for estimation of porosity from density log. In a dolomite formation, Matrix density = 2.87, Density log = 2.44, and Density Fluid = 1.0 (Freshwater) . Solve for Porosity = ? (CO4)

5. (a) (i) Write the operating principle of bore hole compensated Sonic (BHC) logging tools. (CO5)

(ii) Write the value of sonic log at the following depths :

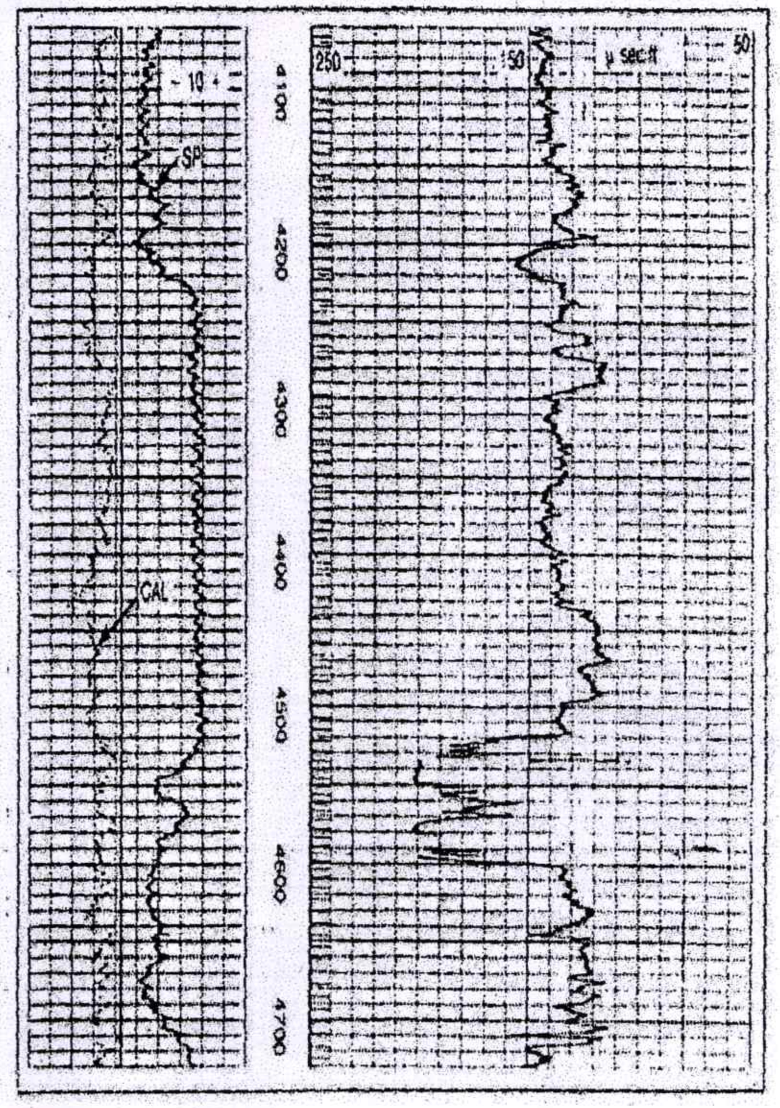
4200 m

4300 m

4400 m

4500 m

4600 m



(b) Derive Wyllie time-average equation for prediction of porosity estimation. Solve this equation if Δt is 81 us/ft and the matrix $\Delta t_{ma} = 55.5$ us/ft and $\Delta t_{fluid} = 189$ us/ft. (CO5)

(c) Explain cross-plot techniques for identification of lithology and mineralogy ? Describe Hi-tech logging tools like, CMR, DSI and FMI.

(CO6)

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Roll No. ....

TPE–503

B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

PETROLEUM PRODUCTION ENGG.—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is composition of petroleum and elemental composition of crude oil by weight percentage ? Write classification of crude oil. (CO1,2)

   (b) Illustrate with the help of a diagram emulsion and types of emulsion. How the emulsions are formed ? Explain surface active agents. (CO1, 2)

   (c) Write a short note on how produced water is treated in Petroleum Industry. (CO1, 2)

2. (a) Illustrate well head. Why they are used in Petroleum Industry. Write its functions. (CO2)

P. T. O.

(b) With the help of Amine treater or sweater explain the process of Gas sweetening and also explain Glycol dehydration with the help of TEG. (CO2)

(c) Explain demulsification. What are the different emulsion treating methods ? (CO2)

3. (a) Explain tank breathing system and tank battery. What do you understand by gauge hatches ? Explain. (CO2, 3)

(b) Differentiate clearly between a two-phase separator and a three-phase separator and write its uses in petroleum industry. (CO2, 3)

(c) Describe what are the different methods by which we can control liquid leaks from tank and how leaks will be identified. (CO2, 3)

4. (a) Write the advantages and limitation of reciprocating separable compressors and reciprocating integral compressors. (CO4)

(b) Differentiate among positive displacement, dynamic and reciprocating compressors. (CO4)

(c) What is a gas meter ? Describe different types of gas meter. (CO4)

5. (a) Explain what is water injection and describe the sources of injected water. (CO3,5)

(b) Explain fixed roof tank and floating roof tank and justify why Floating roof tank are best for petroleum industry. Differentiate between external floating roof tank and internal floating roof tank. (CO3, 5)

(c) Write a short note on gas blanketing system and early production facility or system. (CO3, 5)

Roll No. ....

TPE–504

B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

RESERVOIR ENGINEERING—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Using diffusivity equation prove dimensionless pressure. (CO1, 2)

(b) Determine the pressure response at the wellbore due to the production of a well which is situated at the centre of a square drainage area with sides measuring : (CO1, 2)

(i) L= 200 ft

(ii) L = 600 ft

P. T. O.

The relevant reservoir and fluid properties are as follows :

- $k = 65$ mD,
- Porosity = 0.7,
- Viscosity = 1 cp,
- $c = 15 \times 10^{-6}$ psi,
- $r_w = 0.3$ ft
- CA = 30.9

(c) What is non-Darcy flow ? Explain with the help of Darcy law for radial flow. (CO1, 2)

2. (a) Illustrate the constant terminal rate solution for the flow of a real gas. (CO1, 2)

(b) Define A gas reservoir has the following characteristics : (CO1, 2)

- A-3200 acres
- H-32 ft
- ϕ-0.25
- Swi-20%
- T-150° F
- Pi-2600 psi

p	z
2600	0.82
1000	0.88
400	0.92

Calculate cumulative gas production and recovery factor at 1000 and 400 psi.

(c) With the help of a graph explain the concept of Constant Terminal Rate Solution for semi steady state. (CO1, 2)

3. (a) Write a short note on Water influx by describing water influx, occurrences and causes. (CO2, 3)

   (b) Describe Gas condensate and Saturated oil reservoirs. (CO2, 3)

   (c) (i) Show, using dimensional analysis, that both tD, and pD are dimensionless. (CO2, 3)

   (ii) Express tD, in field units with the real time in hours and days.

4. (a) Write 9 Typical measures of pressure in Petroleum Reservoirs. (CO3, 4)

   (b) Explain Reservoir Pressure Measurement and Reservoir Management Process. (CO3, 4)

   (c) Write short notes on any *two* of the following : (CO3, 4)

   (i) Surface Tension

   (ii) Rock and fluid compressibilities

   (iii) Generalized MBE Equation

5. (a) What is drive mechanism in reservoir engineering ? Explain in detail combination drive and which drive mechanism gives highest recovery ? (CO4, 5, 6)

   (b) What is general theory of well testing ? Explain. (CO4, 5, 6)

   (c) Describe in brief oil reservoir performance prediction with all its phases. (CO4, 5, 6)

Roll No. ....

TPE–505

B. TECH. (PETROLEUM ENGINEERING) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

PETROLEUM REFINING ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Classify the different physical and chemical processes involved in the refining of crude oil refining. Explain any *one* physical process with process description. (CO1, CO2)

(b) What do you mean by quality control of the petroleum products ? Hence define the following : (CO1, CO2)

(i) Smoke point

(ii) Cloud point

(iii) Electric desalting

(iv) Pour point

P. T. O.

(v) Fire point

(vi) Octane number

(vii)Cetane number

(viii) Viscosity index

(c) What is fractionation distillation ? Draw the flowsheet diagram of Atmospheric Distillation Unit (ADU) and discuss its process description with stepwise. (CO3)

2. (a) Discuss all the five basic areas of Petroleum refining processes and operations. Explain in brief of all the operations. (CO3, CO4)

(b) Tabulate the physical and chemical processes involved in petroleum refining. Explain the types of trays and plates in the distillation column. (CO1, CO4)

(c) Design the flow sheet of aromatic solvent extraction unit. Explain solvent extraction. What are the widely used extraction solvents ? What are the purpose of solvent extraction ? (CO1, CO5)

3. (a) What do you mean by thermal process in the refining of crude oil ? Discuss Vis breaking with the flowsheet diagram and illustrate the process description. (CO1)

(b) Why do we use delayed coking ? What are the three typical types of coke obtained ? Explain the steps only during the delayed coking. (CO1, 2)

(c) What are the different catalytic processes ? How cracking is catalyzed ? What are the other reactions during the catalytic cracking. Explain any *one* of them. (CO2, CO5)

4. (a) What do you mean by FCC ? Design the flowsheet diagram of detailed FCC unit and explain their process description. (CO1, CO5)

   (b) Define the process hydro treating. Why do we hydro treating ? Illustrate catalytic hydrodesulphurization in detail with the help of trickle bed rector diagram. (CO4, CO5)

   (c) What is hydrocracking ? Draw the different hydrocracking configurations. Explain the process description of hydrocracking.

   (CO1, CO6)

5. (a) Define reformates. Draw and explain the process description of the flow sheet scheme of catalytic reforming. (CO4, CO6)

   (b) What do you mean by alkylation ? Explain the sulfuric acid alkylation with the suitable flowsheet diagram. (CO4, CO5)

   (c) Explain fluid coking and flexicoking. What are the different reactors used for catalytic hydrogenation ? Explain them with the labelled diagram. (CO2, CO3)

Roll No. ....

TPE–512

B. TECH. PETROLEUM ENGINEERING

(FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2023

OIL FIELD INSTRUMENTATION AND CONTROL

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you mean by calibration of an instrument ? What are the calibration procedure and how it can be classified ? Discuss the different types of errors in details. (CO1, CO2)

   (b) Explain the following : (CO1, CO2)

   (i) Instrumentation

   (ii) Process Instrumentation

   (iii) Transducer

   (iv) Mechanical Amplifier

   (v) Hydraulic Amplifier

   (c) Discuss classification of temperature sensors based on Mechanical sensor, Electrical sensor and Radiation sensors. (CO3)

P. T. O.

2. (a) Describe RTD and its material ? Explain Resistance Temperature Detector (RTD) with their principle and working. (CO3, CO4)

   (b) Explain for the following principles of thermoelectric temperature measurements : (CO1, CO4)

   (i) Seebeck effect

   (ii) Peltier effect

   (iii) Thompson effect

   (c) Discuss the classification of liquid level measuring methods ? Discuss direct level measurement in detail. (CO1, CO5)

3. (a) Discuss and explain in liquid level measurements : (CO1)

   (i) Hydrostatic method

   (ii) Radiation method

   (iii) Ultrasonic level detector

   (b) Define the flow measurement. What are the different methods of flow measurements ? (CO1, CO2)

   (c) Discuss the working of orifice flow meter and hence derive and flow equations of orifice meter. (CO2, CO5)

4. (a) Define Laplace transform ? Why the uses of Laplace transform necessary in control system ? Illustrate the properties of Laplace transform. (CO1, CO5)

(b) Solve for $x(t)$, the equation given as : (CO4, CO5)

$$\frac{d^3x}{dt^3}+2\frac{d^2x}{dt^2}-\frac{dx}{dt}-2x=4+e^t$$

$$x(0)=1$$

$$x'^{(0)}=0,\ x''^{(0)}=-1$$

(c) Derive an expression for the transfer function first-order in series for interacting system specify the symbol used. (CO1, CO6)

5. (a) What derive an expression for transfer function of first order mixing process ? (CO4, CO6)

$$\frac{Y(s)}{X(s)}=\frac{1}{\tau(s)+1}$$

Define τ for the same.

(b) Drive an expression for transfer function liquid level first order system; also define the terms and symbols used. (CO4, CO5)

(c) Write notes on instrumentation and control in separation and oil-well drilling operations. (CO2, CO3)

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